DFD6760

Fully Automatic Dicing Saw

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High end dicing saw for Φ300 mm wafers

Φ300 mm　Facing dual spindle



Fully automatic dicing saw with dual chuck tables

DFD6760 is a high end model which has further enhanced productivity from the existing model, DFD6362. Since this dicing saw has two chuck tables (hereafter called C/T), it is possible for transfer and alignment to occur on one chuck table, while processing is performed on the other chuck table. This shortens the spindle waiting time for processing considerably and improves productivity. In addition, inspection and measurement operations including processed workpiece image recording and workpiece thickness measurement using an NSD* can be performed on one chuck table, while processing proceeds on the other chuck table to maintain throughput. Increased X, Y and Z-axis feed speeds (same as the existing model) also contribute to the reduced processing time.

* Non-contact Surface Detector （option）

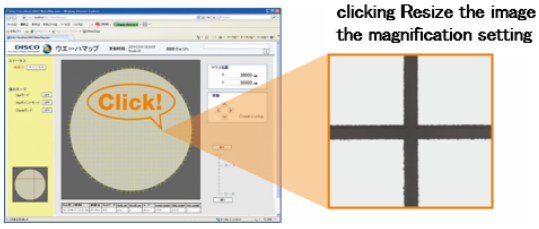
Enhancing the quality of thin wafer processing while maintaining throughput

For thin wafer step cuts, controlling the cut depth from the wafer surface is important. The NSD that is used to control the cut depth makes it possible to measure the wafer upper surface height very accurately. Typically, the NSD requires about 10 seconds to make the measurements, but in the case of the DFD6760, the NSD makes the measurements while processing the wafer on the other chuck table, so the throughput does not deteriorate.

Meeting the need for traceability

Since the DFD6760 is equipped with two chuck tables, while dicing is performed on one chuck table, kerf images of the workpiece after processing can be saved and processing quality can be recorded using the other chuck table. In recent years at production facilities, there has been a demand for traceability. If a defect is found in a device, the DFD6760 can provide the records to check or certify product quality.

Dicer image viewer and software for browsing processing records



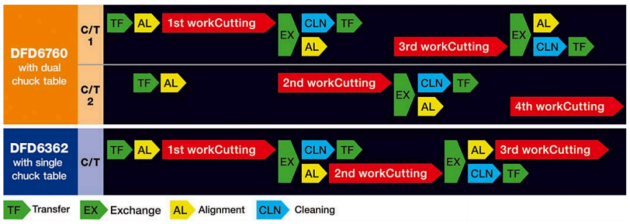
Various measures to prevent particles

The cut section atomizing nozzle, wheel cover, and chuck table water curtain, which have been well received in the DFD6362, are selectable as optional accessories in accordance with workpiece characteristics.

Easy operation

The GUI (Graphical User Interface) implemented on the DFD6362, which has been recognized for its ease of use, has been included on the DFD6760. This is combined with an LCD touch panel for stress-free operation.

Process flow comparison with existing model


DFD6760
with dual chuck table
C/T 1
TF AL 1st workCutting EX CLN TF AL 3rd workCutting EX AL CLN TF
C/T 2
TF AL 2nd workCutting EX CLN TF AL 4th workCutting
DFD6362
with single chuck table
C/T
TF AL 1st workCutting EX CLN TF AL 2nd workCutting EX AL 3rd workCutting CLN TF
TF Transfer
EX Exchange
AL Alignment
CLN Cleaning

Specifications

Specification		Unit	1.2, 1.8 kW	2.2 kW
Max. workpiece size		mm	Φ300	
X-axis	Cutting range	mm	310	
	Cutting speed	mm/s	0.1 ~ 1,000	
Y1・Y2-axis	Cutting range	mm	310	
	Index step	mm	0.0001	
	Positioning accuracy	mm	Within 0.003/310 (Single error) Within 0.002/5	
MY1・MY2-axis	Index step	mm	0.0001	
	Positioning accuracy	mm	Within 0.003/310 (Single error) Within 0.002/5	
Z-axis	Max. stroke	mm	14.7 (For Φ2 inch blade)	14.9 (For Φ3 inch blade)
	Moving resolution	mm	0.00005	
	Repeatability accuracy	mm	0.001	
θ-axis	Max. rotating angle	deg	380	
Spindle	Rated torque	N・m	0.19 (1.2 kW/High-speed rotation) 0.29 (1.8 kW)	0.7
	Rotation speed range	min^{-1}	6,000 ~ 60,000 (1.2 kW/1.8 kW) 20,000 ~ 80,000 (High-speed rotation)	3,000 ~ 30,000
Equipment dimensions(W×D×H)		mm	1,200 × 1,900 × 1,800	
Equipment weight		kg	Approx. 2,750	

*Product appearance, features, specifications, and other details may change due to technical modifications.
*Please read the standard specification sheet thoroughly before use.

Product Lineup







	DFD6240	DFD6340	DFD6341	DFD6342	DFD6361
Machine type	Fully Automatic	Fully Automatic	Fully Automatic	Fully Automatic	Fully Automatic
Spindle	1	2	2	2	2
C/T	1	1	1	1	1
Machine dimentions (W × D × H)	900 × 1190 × 1800	1180 × 1110 × 1850	1180 × 1080 × 1820	1180 × 1080 × 1800	1200 × 1550 × 1800 1
Machine weight (kg)	1200	1600	1500	1240	2050

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